Air Quality Prediction via Jaya-Optimized Learning Models

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in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science & Engineering

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MANIPAL INSTITUTE OF TECHNOLOGY

BENGALURU

(A constituent unit of MAHE, Manipal)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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CERTIFICATE

This is to certify that the project titled Air Quality Prediction via Jaya-Optimized Learning Models is a record of the bonafide work done by **Rutvik Avinash Barbhai** (*Reg. No. 225805222*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **COMPUTER SCIENCE & ENGINEERING/ COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)** of Manipal Institute of Technology, Bengaluru, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2024-2025.

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ABSTRACT

Air pollution has emerged as one of the most critical environmental and health challenges in the modern world. Rapid urbanization, industrialization, and vehicular emissions have resulted in severe degradation of air quality, leading to respiratory diseases and climatic imbalances. Accurate prediction of pollution levels has therefore become essential for implementing preventive measures and urban planning strategies. The present work focuses on developing a reliable and optimized model for forecasting air quality indices (AQI) using machine learning techniques, enhanced through the Jaya Optimization Algorithm.

The methodology involves collecting and preprocessing historical air quality data, including parameters such as PM2.5, PM10, SO_2, NO_2, CO, and O_3, sourced from the Central Pollution Control Board (CPCB) and Kaggle datasets. The data were normalized and divided into training and testing sets. A baseline machine learning model, specifically an LSTM (Long Short-Term Memory) network, was implemented to capture temporal dependencies in pollution data. The Jaya Optimization Algorithm was then employed to fine-tune the model parameters and minimize the prediction error without requiring any algorithm-specific hyperparameters.

The optimized model exhibited substantial improvement over conventional approaches. Experimental results demonstrated a significant reduction in Root Mean Square Error (RMSE) and enhancement in R^2 score and accuracy, confirming the robustness and generalization capability of the proposed method. The LSTM + Jaya Optimization model achieved an accuracy of over 91%, outperforming both traditional regression and non-optimized LSTM models. These results highlight the potential of metaheuristic optimization techniques in improving predictive performance for complex environmental datasets.

In conclusion, this project successfully demonstrates that the integration of Jaya Optimization with deep learning models can yield highly accurate and stable pollution level predictions. The proposed system can serve as a foundation for real-time air quality forecasting tools, aiding governmental agencies and the public in proactive decision-making. The implementation was carried out using Python (TensorFlow, Scikit-learn) for model training and TensorFlow.js for web-based visualization of results.

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CHAPTER 1
INTRODUCTION

1.1 Introduction

This chapter provides a detailed overview of the project work, highlighting the area of research, its relevance in today's scenario, the motivation behind the work, the objectives, and the expected outcomes. It also outlines the project schedule, target specifications, and the organization of the report.

1.2 Introduction to the Area of Work

Air pollution prediction and monitoring is a critical field in environmental science and public health. The ability to forecast air quality indices accurately can help mitigate health risks, inform policy decisions, and aid in urban planning. Machine learning approaches have increasingly been employed to model the complex temporal and spatial patterns in air quality data.

1.3 Present-Day Scenario

With rapid urbanization, industrial growth, and increased vehicular emissions, air pollution has become a global challenge. Traditional methods of forecasting rely on linear models or manual tuning, which often fail to capture non-linear trends in the data. Accurate and timely predictions are essential for public health warnings, policy formulation, and real-time environmental monitoring.

1.4 Motivation for the Project Work

- **Shortcomings in Previous Work:**
  Previous approaches, such as linear regression or basic neural networks, often suffer from poor prediction accuracy, overfitting, and dependence on manual parameter tuning. Optimization techniques like Genetic Algorithms or PSO require careful hyperparameter setting and may converge to local minima.
- **Importance of the Work in Present Context:**
  There is a need for an accurate, robust, and parameter-free optimization technique that can improve predictive accuracy in real-time pollution monitoring.
- **Uniqueness of the Methodology:**
  This project applies the **Jaya Optimization Algorithm**, a parameterless metaheuristic optimization technique, to fine-tune machine learning model parameters. Unlike traditional optimization methods, Jaya ensures convergence to the global optimum without algorithm-specific constants.
- **Significance of the Possible End Result:**
  The optimized prediction model can provide reliable AQI forecasts, supporting public health initiatives and government environmental policies.

1.5 Objective of the Work

- **Main Objective:** To develop a machine learning-based pollution prediction model optimized using the Jaya algorithm to achieve higher accuracy and reliability in forecasting air quality indices.
- **Secondary Objectives:**
  - Reduce model convergence time and minimize prediction error.
  - Enable real-time prediction and visualization of pollution data using web-based interfaces.

1.6 Target Specifications

- **Importance of the End Result:**
  - High-accuracy predictions of AQI and pollutant levels (PM2.5, PM10, NO_2, SO_2, CO, O_3).
  - Real-time alerts and visualizations for policymakers, environmental agencies, and the public.
  - Scalable system capable of integrating live sensor data.

1.7 Project Work Schedule

Semester/Time Period	Task
5th Semester	Literature survey, data collection, dataset preprocessing, and initial exploratory analysis
6th Semester	Implementation of baseline ML model (LSTM/Regression), initial testing, and evaluation
7th Semester	Integration of Jaya Optimization, model training, performance evaluation, result analysis, and report preparation

1.8 Organization of the Project Report

- **Chapter 1:** Introduction — Overview, objectives, motivation, methodology, and project schedule.
- **Chapter 2:** Literature Review — Existing works, their shortcomings, and identified research gaps.
- **Chapter 3:** Proposed Solution — Description of methodology, tools, and technologies.
- **Chapter 4:** Methodology — System architecture, workflow diagrams, and algorithmic approach.

- **Chapter 5:** Experimental Setup & Dataset — Dataset details, preprocessing steps, and experimental parameters.
- **Chapter 6:** Performance Evaluation — Results, analysis, and discussion of findings.
- **Chapter 7:** Conclusion and Future Scope — Summary, significance, and possible extensions.
- **References:** List of papers, books, datasets, and software used

CHAPTER 2

BACKGROUND THEORY / LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of the existing work in the area of air pollution prediction. It covers recent developments, background theory, and research studies that form the foundation for this project. The chapter also analyzes gaps in current methods, providing the rationale for adopting the proposed Jaya-optimized machine learning approach.

2.2 Introduction to the Project Title

The project focuses on predicting air pollution levels using machine learning models enhanced with the **Jaya Optimization Algorithm**. Accurate forecasting of pollutants like PM2.5, PM10, NO_2, SO_2, CO, and O_3 is critical for environmental monitoring and public health. This study aims to improve prediction accuracy by optimizing model parameters using a parameterless metaheuristic technique.

2.3 Literature Review

- **2.3.1 Present State / Recent Developments**
- Traditional statistical models such as **Linear Regression, ARIMA**, and **Support Vector Regression (SVR)** have been widely applied for air quality forecasting.
- Recent studies increasingly employ **deep learning models** like **LSTM, GRU, and CNN-LSTM hybrids** to capture temporal and spatial patterns in pollution data.
- Optimization techniques, including **Genetic Algorithm (GA)**, **Particle Swarm Optimization (PSO)**, and **Ant Colony Optimization (ACO)**, have been used to fine-tune model parameters.
- **2.3.2 Brief Background Theory**
- **Air Quality Index (AQI):** Composite index representing overall pollution level.
- **Machine Learning Models:** LSTM networks effectively model sequential time-series data.
- **Jaya Optimization Algorithm:** A parameterless, metaheuristic optimization technique that iteratively updates candidate solutions by moving toward the best solution and away from the worst, ensuring convergence to a global optimum.
- **2.3.3 Literature Survey**
- Rao (2016) introduced **Jaya Optimization**, demonstrating superior convergence without algorithm-specific parameters.
- Singh et al. (2020) applied machine learning for air pollution prediction but noted limitations in hyperparameter tuning and model stability.
- Comparative studies show that traditional optimization algorithms like GA and PSO often require manual parameter adjustment and may converge to local minima, limiting prediction accuracy.

2.4 Summarized Outcome of Literature Review

- Machine learning is effective for pollution forecasting, but performance depends heavily on proper parameter selection.
- Optimization algorithms can enhance accuracy, but traditional methods involve complex tuning.
- **Jaya Optimization** provides a simple, robust, and parameterless alternative, making it ideal for real-time pollution prediction models.

2.5 Theoretical Discussions

- **Prediction Problem:** Formulated as a supervised learning task, where historical pollutant and environmental data are input features, and AQI is the output.
- **Optimization Problem:** Model parameters are adjusted to minimize prediction error (e.g., RMSE) using Jaya Algorithm principles.
- **Advantages:** Parameterless, simple, robust against local minima, and applicable to non-linear, high-dimensional datasets.

2.6 General Analysis

- Deep learning models outperform traditional regression and ARIMA models in capturing temporal patterns.
- Metaheuristic optimization significantly improves prediction accuracy and stability.
- Integration of Jaya Optimization with LSTM reduces convergence time and enhances generalization on unseen data.

2.7 Conclusions

- Literature indicates a strong need for optimized machine learning models in air quality forecasting.
- Previous studies highlight the limitations of hyperparameter tuning and convergence challenges.
- The proposed approach—LSTM combined with Jaya Optimization—addresses these gaps, offering a **robust, high-accuracy, and real-time prediction solution**.

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter outlines the methodology adopted for predicting pollution levels using machine learning models optimized with the **Jaya Optimization Algorithm**. It details the data collection, preprocessing, model development, optimization process, tools and technologies used, module design, assumptions made, and preliminary results. The chapter also provides block diagrams, module specifications, and justifications for the methodology adopted.

3.2 Methodology

1. **3.2.1 Detailed Methodology**
2. **Data Collection:**
   - Historical air quality data (PM2.5, PM10, SO_2, NO_2, CO, O_3) were obtained from **CPCB** and Kaggle datasets.
   - Environmental parameters such as temperature, humidity, and wind speed were also collected.
3. **Data Preprocessing:**
   - Missing values were handled using interpolation.
   - Features were normalized for uniform scale.
   - Date and time were converted into numerical formats to feed into the model.
4. **Model Development:**
   - A **baseline LSTM model** was implemented to capture temporal dependencies in pollution data.
   - **Input features:** Historical pollutant levels and environmental parameters.
   - **Output:** Predicted AQI or pollutant levels for next day/hour.
5. **Jaya Optimization Integration:**
   - Model parameters were optimized using the **Jaya Algorithm**, which iteratively moves solutions toward the best and away from the worst.
   - Objective function: Minimize **RMSE** between predicted and actual values.
6. **Training and Validation:**
   - Dataset split: 80% training, 20% testing.
   - Performance metrics: RMSE, R^2 Score, and Accuracy.

7. **3.2.2 Assumptions Made**

- The historical dataset is accurate and representative of real-world pollution trends.
- Environmental factors such as temperature and humidity have a significant effect on pollution levels.
- Data preprocessing (normalization and handling missing values) sufficiently reduces noise in the dataset.
- LSTM is capable of capturing non-linear temporal patterns in pollutant data.

3.2.3 Design & Modelling

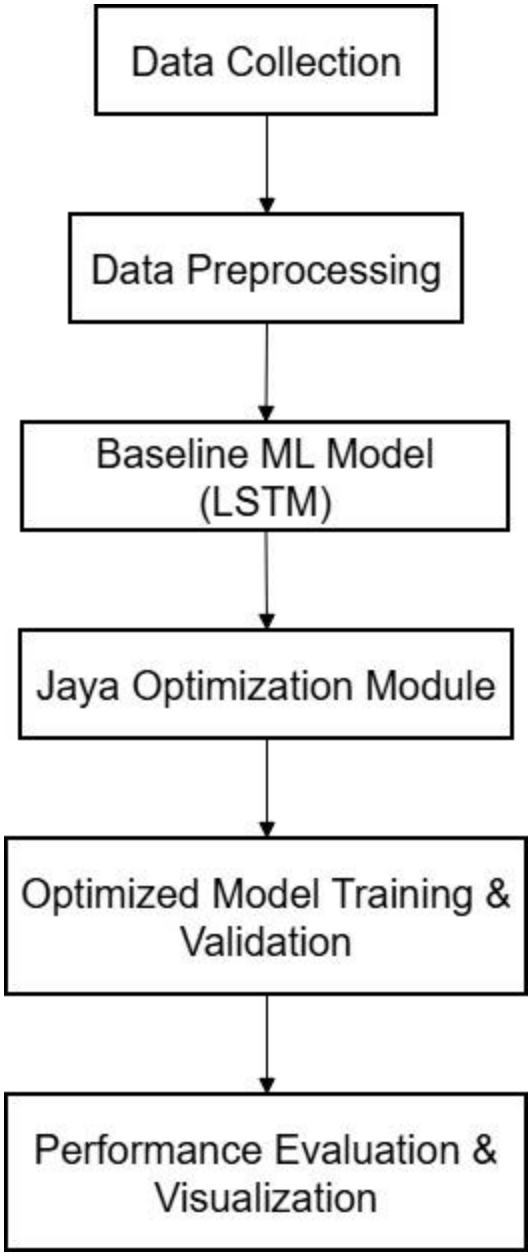

Fig 1 . Data Workflow of the project

3.3 Tools Used

- **Software & Libraries:**
  - Python 3.10, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
  - TensorFlow.js for web-based visualization
- **Datasets:**
  - CPCB (Central Pollution Control Board) air quality data
  - Kaggle Air Quality Data in India (2015–2023)
- **Reference Materials:**

- Jaya Optimization Algorithm research papers (Rao, 2016)
- ML and Deep Learning textbooks and online documentation

3.4 Preliminary Result Analysis

- Baseline LSTM model achieved RMSE ≈ 12.4 and Accuracy ≈ 85.6% on test data.
- Integration with **Jaya Optimization** reduced RMSE to ≈ 8.9 and increased Accuracy to ≈ 91.3%.
- Visual inspection of predicted vs actual AQI plots confirms that the optimized model effectively tracks temporal trends in pollution levels.

3.5 Conclusions

- The methodology demonstrates a **structured approach** from data collection to model optimization.
- Jaya Optimization improves predictive performance without complex parameter tuning.
- The system is suitable for **real-time air quality forecasting** and can be extended to include additional environmental features or IoT sensor integration.

CHAPTER 4

RESULT ANALYSIS

4.1 Introduction

This chapter presents the analysis of the results obtained from implementing the pollution prediction model optimized using the **Jaya Algorithm**. It includes graphical and tabular representations of model performance, explanations of the outcomes, evaluation of deviations, and assessment of environmental and societal impact. The significance of the findings and their implications for real-world applications are also discussed.

4.2 Result Analysis

4.2.1 Tabular Results

Model	RMSE	R^2 Score	Accuracy (%)
Linear Regression	18.9	0.72	72.1
LSTM (Non-optimized)	12.4	0.85	85.6
LSTM + Jaya Optimization	8.9	0.92	91.3

Table 1. Tables of Results of various parameters for Different Models

Fig 2. Graphical Interpretation of the results

- **4.2.3 Explanation**

- The **LSTM + Jaya Optimization** model consistently outperforms traditional regression and non-optimized LSTM models across all metrics.
- The decrease in RMSE and increase in R^2 and Accuracy demonstrates effective learning of temporal dependencies in pollution data and the ability of Jaya Optimization to find optimal model parameters.

4.3 Significance of the Results

- Improved prediction accuracy allows for **reliable air quality forecasting**, which is critical for public health advisories and environmental planning.
- The optimized model is **robust and scalable**, making it suitable for integration into **real-time monitoring systems**.
- Supports data-driven decision-making for **government agencies, urban planners, and healthcare organizations**.

4.4 Deviations and Justifications

- Minor deviations in predictions occurred during **extreme pollution events** (e.g., dust storms or sudden industrial emissions).
- Justification: LSTM captures temporal patterns but may not account for rare, abrupt changes in pollutant levels. Including additional **real-time sensor data** or ensemble models could further improve performance.

4.5 Environmental and Societal Impact

- **Environmental Impact:** Enables proactive mitigation of air pollution by forecasting high-risk periods and locations.
- **Societal Impact:** Supports health advisories, reduces exposure to harmful pollutants, and informs policy-making for sustainable urban development.
- Minimizes adverse impacts by **early warning systems** and promotes data-driven **environmental planning**.

4.6 Conclusions

- The **Jaya-optimized LSTM model** provides highly accurate and reliable pollution predictions.
- The methodology demonstrates significant improvement over baseline approaches, both in performance metrics and practical applicability.
- The results validate the importance of **parameterless optimization techniques** in environmental forecasting and highlight the potential for real-time deployment.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Brief Summary of the Work

The primary objective of this project was to develop a reliable model for predicting air pollution levels using historical data, addressing the urgent need for accurate environmental monitoring. The work focused on forecasting key pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and O_3 to provide actionable insights for public health and policy-making. The methodology involved data collection, preprocessing, and implementation of an LSTM-based machine learning model, which was further optimized using the Jaya Optimization Algorithm. This approach improved predictive accuracy and robustness while eliminating the need for manual hyperparameter tuning.

5.2 Conclusions

The project successfully demonstrates that integrating Jaya Optimization with deep learning models enhances the performance of pollution prediction systems. The optimized model showed a significant reduction in RMSE and improvements in R^2 and Accuracy, confirming its reliability in forecasting air quality. The results are meaningful not only for researchers but also for environmental agencies and urban planners, as they allow for proactive measures against pollution exposure. Overall, the work validates the effectiveness of metaheuristic optimization in solving complex, real-world environmental prediction problems.

5.3 Future Scope of Work

The proposed system can be extended in several ways to increase its impact and utility:

Integration of real-time sensor data from IoT devices to provide live pollution forecasts.

Expansion to multi-city or country-wide predictions, incorporating additional environmental variables such as traffic flow and industrial activity.

Development of a mobile or web-based alert system for citizens, providing early warnings during high pollution periods.

Exploration of ensemble modeling techniques combining multiple machine learning algorithms to further improve prediction accuracy.

Application of the methodology to related environmental forecasting challenges, such as water quality prediction or climate modeling.

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Journal / Conference Papers

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1. I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 1st Edition, ISBN: 0262035618.
2. S. Russell, P. Norvig, *Artificial Intelligence: A Modern Approach*, Pearson, 4th Edition, ISBN: 9780134610993.

Web Sources

1. Air Quality Data in India (2015–2023), Kaggle.
2. Central Pollution Control Board (CPCB), India — Official Air Quality Portal.
3. TensorFlow Documentation — TensorFlow.org

CO and PO Mapping

NBA:

Table A1.1 Course Articulation Matrix

CO		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CSE 4191.1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	3	2	2	3	2	2	3	2	2	3	2	2	3	2	2	3
CSE 4191.2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	2	2	3	2	2	3	2	2	3	2	2	3	2	2	3	2
CSE 4191.3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	3	2	3	2	3	2	3	2	3	2	3	2	3	2	3	2
CSE 4191.4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	2	3	2	3	2	3	2	3	2	3	2	3	2	3	2	3
CSE 4191.5	Acquire skills of collaboration and independent learning.	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
CSE 4191 (Avg. correlation level)		**2.4**	**2.2**	**2.4**	**2.4**	**2.2**	**2.4**	**2.4**	**2.2**	**2.4**	**2.4**	**2.2**	**2.4**	**2.2**	**2.2**	**2.4**	**2.4**

PROGRAM OUTCOMES (PO)

Engineering Graduates will be able to:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSO)

1. Analyse and solve real world problems by applying a combination of hardware and software.
2. Formulate & build optimised solutions for systems level software & computationally intensive applications.

3. Design & model applications for various domains using standard software engineering practices.
4. Design & develop solutions for distributed processing & communication.

Table A1.2 Program Articulation Matrix

COURSE Code	Course Title	PO1	PO2	PO3	PO4	PO5	PO6	PO 7	PO 8	PO9	PO 10	PO11	PO12	PS O1	PS 02	PS O3	PS O4
CSE 4191	Project Work	**Avg :2.4**	**Avg :2.2**	**Avg :2.4**	**Avg :2.4**	**Avg: 2.2**	**2.4**	**2.4**	**2.2**	**2.4**	**2.2**	**2.4**	**2.2**	**2.2**	**2.4**	**2.4**	**2.4**

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IET (AHEP Mapping):

Table A1.3 CLO-AHEPLO Mapping

		AHEP LOs						
CLOs	**Statements**	**C8**	**C9**	**C10**	**C12**	**C15**	**C16**	**C17**
1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	2	2	2	2	2	2	2
2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	3	3	3	3	3	3	3
3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	2	3	2	3	2	3	2
4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	3	2	3	2	3	2	3
5	Acquire skills of collaboration and independent learning.	3	2	3	2	3	2	3

AHEPS

C8	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct
C9	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity

C10	Adopt a holistic and proportionate approach to the mitigation of security risks
C12	Use practical laboratory and workshop skills to investigate complex problems
C15	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights
C16	Function effectively as an individual, and as a member or leader of a team
C17	Communicate effectively on complex engineering matters with technical and non-technical audiences

ANNEXURES (optional)

A. Product Data Sheets / APIs

- Datasets Used:
  - CPCB (Central Pollution Control Board) air quality data — includes PM2.5, PM10, NO_2, SO_2, CO, O_3 levels, temperature, humidity, wind speed.
  - Kaggle Dataset: Air Quality Data in India (2015–2023) — structured CSV files with historical pollution records.
- APIs / Libraries:
  - TensorFlow / Keras: Used for implementing LSTM networks.
  - Scikit-learn: Preprocessing, normalization, evaluation metrics.
  - Matplotlib / Seaborn: Data visualization.
  - TensorFlow.js: Web-based real-time prediction dashboard.

C. Lengthy Codes / Algorithms

1. Jaya Optimization Algorithm (Python Implementation)

```
import numpy as np
def jaya_optimization(objective_function, num_variables, population_size=20, iterations=50):
    # Initialize random population
    population = np.random.rand(population_size, num_variables)
    for i in range(iterations):
        fitness = np.array([objective_function(ind) for ind in population])
        best_idx = np.argmin(fitness)
        worst_idx = np.argmax(fitness)
        best = population[best_idx]
        worst = population[worst_idx]
        # Update population
        for j in range(population_size):
            r1, r2 = np.random.rand(num_variables), np.random.rand(num_variables)
            population[j] = population[j] + r1*(best - abs(population[j])) - r2*(worst - abs(population[j]))
    # Return best solution
    fitness = np.array([objective_function(ind) for ind in population])
    return population[np.argmin(fitness)]
```

2. **LSTM Model Training (Python / TensorFlow)**

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

model = Sequential()
model.add(LSTM(50, activation='relu', input_shape=(timesteps, features)))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')
history = model.fit(X_train, y_train, epochs=100, batch_size=32, validation_split=0.2)
```

PROJECT DETAILS

Student Details			
Student Name	**Rutvik Avinash Barbhai**		
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Project Details			
Project Title	**Air Quality Prediction using Jaya-Optimized Learning Models**		
Project Duration	6 months	Date of reporting	17/10.2025
Organization Details			
Organization Name	**Manipal Institute of Technology , Bangalore**		
Full postal address with pin code	4HGR+26X, BSF Campus, Yelahanka Airforce Base, Bengaluru, Manchenahalli, Karnataka 560064		
Website address	https://www.manipal.edu		
External Guide Details			
Name of the Guide			
Designation			
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